
Figure 5: Lasagne design principles

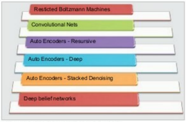


Figure 6: DL4J neural networks

Theano was developed at the Montreal Institute of Learning Algorithms and is available with a BSD licence. Python programmers can solve unique problems with Theano, which cannot be done with Numpy of Python. The following are some examples:

- Theano expressions can utilise both CPU and GPU; they run considerably faster than native Python algorithms.
- Theano has a unique feature of identifying unstable mathematical expressions, and it employs more stable algorithms while evaluating such expressions.

Some of the features of Theano are illustrated in Figure 3.

Many research groups have adopted Theano and there are libraries developed on top of Theano as listed below:

- *Keras:* This is a minimalistic library written in Python on top of Theano, which is highly suitable for quicker prototyping, as well as seamless execution in the CPU and GPU.
- *Pylearn2:* This is also a library that can be used on top of Theano. The developers can build their own Pylearn2 plugins with customised algorithms. The Pylearn2 code is compiled targeting either the CPU or GPU, using Theano.
- *Lasagne:* This is another Theano based library, which is built with six design principles as shown in Figure 4.
- *Blocks:* The blocks framework enables developers to build neural network models on Theano. This framework is used along with Fuel, which provides standards for making machine-learning datasets and enables easier navigation through them.

Theano is more suitable for scenarios where finer control is needed and which require extensive customisation.

## Caffe

Caffe is a super-speciality framework developed to increase the speed of deep learning implementations. Caffe was developed at Berkeley Vision and Learning Centre and is available with a BSD 2-Clause licence. Listed below are some reasons to choose Caffe:

- The switching between CPU and GPU can be carried out by setting a simple flag.
- Caffe is backed by more than 1000 developers with significant contributions. This active community backing is a big plus point.
- Caffe is superfast. Even on a single NVIDIA K40 GPU, it can infer an image in 1ms.

The popularity of Caffe can be gauged from the fact that within just two years of its release, it has got 600+ citations and 3400+ forks. The Caffe framework primarily focuses on vision, but it can be adopted for other media as well. The detailed installation instructions to configure Caffe in Ubuntu are available at *http://caffe.berkeleyvision.org/installation.html.*

The Caffe demo for image classification is available at *http://demo.caffe.berkeleyvision.org.* A sample screenshot with its classification is shown in Figure 5.

Caffe sample implementations for scene recognition, object recognition, visual style recognition, segmentation, etc, are available. The Embedded Café on NVIDIA Jetson TK1 mobile board enables image inference within 35ms.

## DL4J

DL4J or Deeplearning4J is an open source deep learning library which is targeted at Java/Scala. It can be directly used for deep learning implementations by non-researchers in an easier manner compared to Theano. It allows the development of faster prototyping, and is available under the Apache 2.0 licence. The DL4J use cases are listed below:

- Image recognition
- Speech processing
- Spam filtering
- E-commerce fraud detection, etc

The important features of DL4J are:

- GPU integration
- Robust n-dimensional array handling
- Hadoop and Spark scalability
- ND4J, which is a linear algebra library, and runs 200 per cent faster than Numpy
- Canova, which is a vectorisation tool for machine learning libraries.

The neural network architectures supported by DL4J are shown in Figure 6.

## ConvNetJS

As of 2016, JavaScript is everywhere. Deep learning is no exception. ConvNetJS is an implementation of neural networks in JavaScript. The in-browser demos provided by ConvNetJS are really good. ConvNetJS was developed by Dr Karpathy at Stanford University. A special feature is that it doesn't require any dependencies like compilers, installers and GPUs. ConvNetJS has support for the following:

- Reinforcement learning model
- Classification (SVM) and regression (L2)
- Major neural network modules
- Image processing using Convolutional Networks

An easy-to-follow guide for using ConvNetJS is available at *http://cs.stanford.edu/people/karpathy/convnetjs/started.html.*

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